

# CO-2 AT-1

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Q19. Linear search - Element not present

1. Problem Understanding

The given dataset is :

$A = [17, 29, 43, 56, 68, 71]$

The dataset is unsorted, and the task is to search for the target element 50.

Since, the data is not arranged in ascending or descending order, Binary search cannot be replaced applied. Therefore, each element must be checked sequentially from the beginning until the target element is found or elements are checked.

In this case, the element 50 is not present in the dataset, hence, the search continues until the last element and then concludes that the element is absent.

2. Given

Dataset:  $A = [17, 29, 43, 56, 68, 71]$

Target element (key): 50

No. of Elements:  $n=6$

### 3. Input and output

Input:

dataset = [17, 29, 43, 56, 68, 71]

key = 50

Output:

Element 50 is not found in the dataset

### 4. Method selection

Algorithm used

Linear search (sequential search)

Justification

Linear search is the most suitable algorithm because:

- The dataset is unsorted
- It checks elements one by one from the beginning
- No sorting or preprocessing is required
- It is simple and suitable for small datasets

### 5. Algorithm

Linear search (A, n, key)

for  $i = 0$  to  $n - 1$

if  $A[i] == \text{key}$

return  $i$

return  $-1$

## 6. STEP by STEP solution

### Initial Dataset

Position    1    2    3    4    5    6

Element    17    29    43    56    68    71

STEP-1: compare

$17 \neq 50$  move to next

STEP-2:  $29 \neq 50$     "

STEP-3:  $43 \neq 50$     "

STEP-4:  $56 \neq 50$     "

STEP-5:  $68 \neq 50$     "

STEP-6:  $71 \neq 50$     "

Element 50 is not found

## 7. Comparison Table

| Comparison | Element checked | Result    |
|------------|-----------------|-----------|
| 1          | 17              | Not Found |
| 2          | 29              | Not Found |
| 3          | 43              | Not Found |
| 4          | 56              | Not Found |
| 5          | 68              | Not Found |
| 6          | 71              | Not Found |



### 8. Calculations

#### Comparisons

Total comparisons = 6

Result: Element 50 is absent from the dataset

### 9. Time Complexity Analysis

Best case =  $O(1)$

Average case =  $O(n)$

Worst case =  $O(n)$

### 10. Space Complexity

Space complexity =  $O(1)$

### 11. Verification of the Result

The algorithm checked all six elements in the dataset

17  $\rightarrow$  29  $\rightarrow$  43  $\rightarrow$  50  $\rightarrow$  68  $\rightarrow$  71

The element 50 was not found

Therefore, the algorithm correctly returns

- Element Not Found
- Total comparisons = 6

Hence, the obtained result is verified to be correct.

## 12. Interpretation of the result

Linear Search examined every element because the target value 50 was not present in the dataset. Since the search was unsuccessful, the algorithm had to compare the key with all six elements before concluding that the element was absent.

This shows that when the search is unsuccessful, linear search performs its maximum number of comparisons, making it less efficient for large datasets.

## 13. Result

The element 50 is ~~not~~ present in the dataset

[17, 29, 43, 56, 68, 71]

The ~~linear~~ search algorithm checked all 6 elements before concluding that the element is absent.

The algorithm correctly returned "Element not found", requiring 6 comparisons, with  $O(1)$  space complexity and  $O(1)$ ,  $O(n)$ , and  $O(n)$  time complexity for the best, average and worst cases respectively.